9.13 Loaded Dice

WORKSHEET

The question

STEP I 1999, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q12

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself How would you prove $x + \frac{1}{x} \geq 2$ for $x > 0$? When is there equality?

STAGE 1 The extreme totals

Ask yourself In how many ways can you throw a total of 2? A total of 12? Write their probabilities in terms of the q_i and r_j.

STAGE 2 The middle total

Ask yourself A total of 7 can be made in six ways. Which two of them involve only the faces 1 and 6? Write their probabilities using the equations above.

Pause and try Look at what made the proof work. Making 2 and 12 as likely as 7 forces the 1s and 6s to be fairly likely, and then the "mixed" outcomes (1, 6) and (6, 1) alone make 7 too likely. Figure 9.13.1 (in the full section online) shows the cells involved.

STAGE 3 Two throws of one die

Ask yourself If one die is thrown twice, what is the probability that the two numbers are the same, in terms of the q_j? What is it for a fair die?

Ask yourself Can loading make $\sum q_j^2$ smaller than $\frac{1}{6}$? You know that $\sum q_j = 1$.

What's really going on

A proof with polynomials

Here is a completely different proof, using the generating functions of Section 9.6. Let $Q(x) = q_1 x + \cdots + q_6 x^6$ and $R(x) = r_1 x + \cdots + r_6 x^6$. If all totals had probability $\frac{1}{11}$,

then

$$Q(x)R(x) = \frac{x^2 + x^3 + \cdots + x^{12}}{11} = \frac{x^2}{11}(1 + x + \cdots + x^{10}).$$

Now $Q(x)/x$ is a polynomial of degree 5 with real coefficients, and every real polynomial of odd degree has a real root (its graph goes from $-\infty$ to $+\infty$, or the other way). So the left side has a real non-zero root. But $1 + x + \cdots + x^{10} = \frac{x^{11}-1}{x-1}$, whose roots are the 11th roots of unity other than 1, none of which is real. Contradiction. The same argument shows that no loading of two n-sided dice can make the totals equally likely whenever $n - 1$ is odd, that is, for n even. (For odd n the polynomial argument needs more work, but the AM–GM proof above works for every $n \geq 2$.)

Collision probability

The quantity $\sum q_j^2$ in part (ii), the chance that two independent draws agree, is called the *collision probability* of a distribution. Part (ii) shows that it is smallest for the uniform distribution. It measures how concentrated a distribution is: in ecology, $\sum q_j^2$ is Simpson's index of diversity, where q_j is the proportion of species j in a habitat; in economics, it is the Herfindahl index of market concentration; in cryptography, it measures how guessable a password distribution is. And it explains the remark at the end of Section 9.9: unequal birthday probabilities can only make shared birthdays more likely.

Impossibility proofs

Some of the most famous results in mathematics say that something cannot be done: no fraction squares to 2; no ruler-and-compass construction trisects every angle; no formula in radicals solves every quintic; no algorithm decides every mathematical statement. Each changed its subject, because to prove that *no* method works, you must understand the whole space of methods. The loaded dice problem is a miniature: to rule out every loading, we needed an invariant property (the AM–GM inequality) that every loading must obey.

Try this next

1. **Cauchy–Schwarz.** Use the Cauchy–Schwarz inequality, $(\sum a_j b_j)^2 \leq \sum a_j^2 \sum b_j^2$, with $a_j = q_j$ and $b_j = 1$, to give a one-line proof of part (ii).

2. **Two different dice.** If two *different* loaded dice are each thrown once, can the probability that they show the same number be less than $\frac{1}{6}$? Can it be 0?

3. **Uniform totals with other faces.** Show that two dice *can* give equally likely totals if their faces are numbered differently: try one die numbered $0, 1, 2, 3, 4, 5$ and the other $0, 6, 12, 18, 24, 30$. What totals arise? Why does this not contradict the question?

4 **Three dice.** Can three loaded six-sided dice, with faces 1 to 6, make all totals from 3 to 18 equally likely? (*Hint:* use the totals 3, 8, 13 and 18, and AM–GM for six numbers. Why does the polynomial proof fail this time?)

Further reading

- **J. Michael Steele, *The Cauchy–Schwarz Master Class*** (Cambridge University Press). A beautiful introduction to inequalities, starting from AM–GM.
- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press). Generating functions and their use in proofs like the one above.
- **Ian Stewart, *Galois Theory*** (CRC Press), the opening chapters. The most famous impossibility proof of all, for the quintic.

The full guided solution is in Section 9.13:
https://beyondstep.keithmcnulty.org/ch09/loaded-dice.html